# Mini Paper — 1.1 Processors, Input, Output & Storage — ANSWER SHEET

*Separate answer sheet / mark scheme — keep for marking.*

## Mark scheme

**Q1 — [2] (AO1)** - (a) MAR holds the **address** of the memory location to be read from or written to (1). - (b) MDR holds the **data or instruction** just fetched from, or about to be written to, memory (1). - *Examiner tip:* Keep them separate — MAR = address, MDR = data; swapping them is the most common lost mark here.

**Q2 — [4] (AO1)** - The address held in the **PC is copied to the MAR** (1). - A **read signal is sent on the control bus** and the instruction at that address is fetched along the **data bus into the MDR** (1). - The **PC is incremented** so it points to the next instruction (1). - The contents of the **MDR are copied into the CIR**, ready for decoding (1). - *Examiner tip:* Don't forget the PC increment and the MDR→CIR copy — both are routinely dropped.

**Q3 — [4] (AO1)** - (a) Cache is small, very fast memory on/near the CPU storing frequently/recently used instructions and data (1); the CPU can fetch these without accessing slower main memory (RAM), reducing waiting time (1). - (b) Any **one** clear difference for 2 marks, e.g. L1 is **smaller** than L3 (1) and L1 is **faster** / on the core whereas L3 is slower / often shared between cores (1). - *Examiner tip:* Say cache is "on/near the CPU" — calling it "part of RAM" loses the mark.

**Q4 — [3] (AO2)** - Harvard architecture has **separate memories and buses for instructions and data** (1). - This means an instruction can be **fetched at the same time as data is read/written** (avoids the Von Neumann bottleneck) (1). - For the washing machine this gives **faster, more predictable execution** of its fixed control program, which suits a real-time embedded device (1). - *Examiner tip:* The mark for application is tying "simultaneous access" back to the embedded/real-time scenario, not just defining Harvard.

**Q5 — [3] (AO2)** (max 3) - CISC has a **large set of complex, variable-length instructions**, whereas RISC has a **small set of simple, fixed-length instructions** (1). - In CISC the complexity is in **hardware**; in RISC the complexity is in **software/the compiler** (more instructions per task) (1). - RISC's uniform single-cycle instructions are **easier to pipeline** and use **less power**, whereas CISC instructions may take several cycles (1). - *Examiner tip:* Actually compare using "whereas"/"compared to"; two separate descriptions caps your marks.

**Q6 — [3] (AO2)** (max 3) - A GPU has **many (thousands of) cores** designed for parallel processing (1). - Neural network training applies the **same operation to large blocks of data** (data parallelism), so the independent calculations run **simultaneously** across the cores (1). - A CPU has only a **few cores optimised for sequential tasks**, so it would be slower for this highly parallel workload (1). - *Examiner tip:* Name the data-parallel nature of the work — "GPUs are just faster" without explaining *why* scores poorly.

**Q7 — [3] (AO1/AO2)** - Virtual memory is when **RAM is full** and the OS uses a portion of **secondary storage (a swap/paging file) as if it were extra RAM** (1), moving pages not currently needed out of RAM and back when required (1). - Drawback: secondary storage is **far slower than RAM**, so heavy paging causes **disk thrashing** and slows the system (1). - *Examiner tip:* Don't confuse virtual memory (disk used as overflow RAM) with cloud/virtual storage (remote servers).

**Q8 — [8] (AO3)** (levelled, max 8)

Indicative content — credit reasoned, applied evaluation:

**(a) Laptop drive — recommend flash/SSD:** - **No moving parts → durable and shock-resistant**, important for journalists filming on location (1). - **Fast** access, **low power** (better battery life), **silent** and **lightweight/portable** for travel (1). - Trade-off acknowledged: more expensive per GB and finite write cycles, but capacity is sufficient for a laptop (1).

**(b) Archiving large rarely-accessed footage — recommend magnetic (HDD or tape):** - **Very large capacity at low cost per GB**, ideal for huge volumes of old footage (1). - **Tape** in particular is cheap and reliable for long-term archive; slow **serial access** matters little as the data is rarely retrieved (1). - Trade-off acknowledged: slower with moving parts/fragility, acceptable for infrequent backup access (1).

**Evaluation / comparison (higher band):** - Explicitly weighs **cost vs speed vs durability vs access pattern** for each scenario and reaches a clear justified choice for both (1). - Considers and rejects **optical** with reason (cheap and portable but far too low capacity for large archives) (1).

*Level guide:* L3 (7–8) sustained, well-applied evaluation with justified choices for both purposes; L2 (4–6) some application and comparison; L1 (1–3) basic descriptions with limited application.

*Examiner tip:* For both parts, link each storage characteristic directly to a stated requirement (travel/durability; volume/cost/rare access) — generic pros/cons without application stays at the lower bands.